

## SAFETY DATA SHEET

Revision Date 27-March-2024

Revision Number 3

### 1. Identification

**Product Name** 1,2,3-trichloropropane

**Cat No. :** L04312

**CAS-No** 96-18-4  
**Synonyms** Allyl trichloride; Glycerol trichlorohydrin; Trichlorohydrin

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                              |             |
|------------------------------|-------------|
| <b>Flammable liquids</b>     | Category 4  |
| <b>Acute oral toxicity</b>   | Category 4  |
| <b>Acute dermal toxicity</b> | Category 4  |
| <b>Carcinogenicity</b>       | Category 1B |
| <b>Reproductive Toxicity</b> | Category 1B |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Combustible liquid

Harmful if swallowed or in contact with skin

May cause cancer  
May damage fertility

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF exposed or concerned: Get medical advice/attention  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component              | CAS-No  | Weight % |
|------------------------|---------|----------|
| 1,2,3-Trichloropropane | 96-18-4 | > 99     |

### 4. First-aid measures

|                                        |                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                              |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.                 |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                   |
| <b>Ingestion</b>                       | Clean mouth with water. Get medical attention.                                                                                               |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                        |

### 5. Fire-fighting measures

|                                         |                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Water mist may be used to cool closed containers. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                         |
| <b>Flash Point</b>                      | 74 °C / 165.2 °F                                                                                                                                                                 |
| <b>Method -</b>                         | No information available                                                                                                                                                         |
| <b>Autoignition Temperature</b>         | 304 °C / 579.2 °F                                                                                                                                                                |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                  |
| <b>Upper</b>                            | 12.6%                                                                                                                                                                            |
| <b>Lower</b>                            | 3.2%                                                                                                                                                                             |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                         |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                         |

**Specific Hazards Arising from the Chemical**

Combustible material. Flammable. Vapors may form explosive mixtures with air. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Remove all sources of ignition. Take precautionary measures against static discharges.                          |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system. |

**Methods for Containment and Clean Up** Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition.

## 7. Handling and storage

**Handling** Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Keep away from open flames, hot surfaces and sources of ignition.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong bases. Metals. Finely powdered metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component              | Alberta                                          | British Columbia | Ontario TWA/EV | Quebec         | ACGIH TLV      | OSHA PEL                                                                                                  | NIOSH                                                     |
|------------------------|--------------------------------------------------|------------------|----------------|----------------|----------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1,2,3-Trichloropropane | TWA: 10 ppm<br>TWA: 60 mg/m <sup>3</sup><br>Skin | TWA: 10 ppm      | TWA: 0.005 ppm | TWA: 0.005 ppm | TWA: 0.005 ppm | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 60 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 300 mg/m <sup>3</sup> | IDLH: 100 ppm<br>TWA: 10 ppm<br>TWA: 60 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Viton (R)      | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Physical State                   | Liquid                                     |
| Appearance                       | Light yellow                               |
| Odor                             | sweet                                      |
| Odor Threshold                   | No information available                   |
| pH                               | Not applicable                             |
| Melting Point/Range              | -14 °C / 6.8 °F                            |
| Boiling Point/Range              | 152 - 156 °C / 305.6 - 312.8 °F @ 760 mmHg |
| Flash Point                      | 74 °C / 165.2 °F                           |
| Evaporation Rate                 | No information available                   |
| Flammability (solid,gas)         | Not applicable                             |
| Flammability or explosive limits |                                            |
| Upper                            | 12.6%                                      |
| Lower                            | 3.2%                                       |
| Vapor Pressure                   | 4.5 mbar @ 20 °C                           |
| Vapor Density                    | 5.1 (Air = 1.0)                            |
| Specific Gravity                 | 1.386                                      |

|                                        |                          |
|----------------------------------------|--------------------------|
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 304 °C / 579.2 °F        |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C3 H5 Cl3                |
| Molecular Weight                       | 147.43                   |

## 10. Stability and reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                               |
| Stability                        | Stable under normal conditions.                                                          |
| Conditions to Avoid              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong bases, Metals, Finely powdered metals                                             |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Phosgene, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component              | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation                           |
|------------------------|--------------------------|-----------------------------|-------------------------------------------|
| 1,2,3-Trichloropropane | LD50 = 150 mg/kg ( Rat ) | LD50 = 250 mg/kg ( Rabbit ) | LC50 > 4800 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component              | CAS-No  | IARC     | NTP                    | ACGIH | OSHA | Mexico |
|------------------------|---------|----------|------------------------|-------|------|--------|
| 1,2,3-Trichloropropane | 96-18-4 | Group 2A | Reasonably Anticipated | A2    | X    | A3     |

|                                            |                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Mutagenic Effects                          | No information available                                                                                            |
| Reproductive Effects                       | No information available.                                                                                           |
| Developmental Effects                      | No information available.                                                                                           |
| Teratogenicity                             | Teratogenic effects have occurred in experimental animals.                                                          |
| STOT - single exposure                     | None known                                                                                                          |
| STOT - repeated exposure                   | None known                                                                                                          |
| Aspiration hazard                          | No information available                                                                                            |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component              | Freshwater Algae | Freshwater Fish                                                                                                                                                            | Microtox   | Water Flea                                                                                      |
|------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------|
| 1,2,3-Trichloropropane | Not listed       | LC50: = 75 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 25.9 - 28.9 mg/L, 96h static (Pimephales promelas)<br>LC50: = 50.8 mg/L, 96h flow-through (Pimephales promelas) | Not listed | EC50: 27.8 - 41.1 mg/L, 48h semi-static (Daphnia magna)<br>LC50: = 20 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2810  
Hazard Class 6.1  
Packing Group III

### TDG

UN-No UN2810  
Hazard Class 6.1  
Packing Group III

### IATA

UN-No UN2810  
Proper Shipping Name TOXIC LIQUID, ORGANIC, N.O.S.\*  
Hazard Class 6.1  
Packing Group III

### IMDG/IMO

UN-No UN2810  
Proper Shipping Name Toxic liquid, organic, n.o.s.  
Hazard Class 6.1  
Packing Group III

## 15. Regulatory information

### International Inventories

| Component              | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1,2,3-Trichloropropane | 96-18-4 | X   | -    | X    | ACTIVE                                        | 202-486-1 | -      | -   |

| Component              | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------|---------|-------|----------|------|------|------|------|-------|-------|
| 1,2,3-Trichloropropane | 96-18-4 | X     | KE-34093 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component              | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 1,2,3-Trichloropropane | Part 4 Substance                                     |                                                                            |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component              | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1,2,3-Trichloropropane | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - 202-486-1 - Carcinogenic, Article 57a; Toxic for reproduction, Article 57c      |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

**REACH links**<https://echa.europa.eu/authorisation-list><https://echa.europa.eu/substances-restricted-under-reach><https://echa.europa.eu/candidate-list-table>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component              | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| 1,2,3-Trichloropropane | 96-18-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component              | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1,2,3-Trichloropropane | 96-18-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y45                      |

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**16. Other information**

|                         |                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Product Safety Department<br>Email: chem.techinfo@thermofisher.com<br>www.thermofisher.com |
| <b>Revision Date</b>    | 27-March-2024                                                                              |
| <b>Print Date</b>       | 27-March-2024                                                                              |
| <b>Revision Summary</b> | New emergency telephone response service provider.                                         |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**